SPINK7-KLK5 MD Pipeline: Binding Free Energy via Enhanced Sampling

OpenMM ≥8.1

License MIT

python 3.10+

tests 259 passed

force field AMBER ff14SB

An end-to-end molecular dynamics simulation pipeline for computing the binding free energy of the **SPINK7-KLK5** protease-antiprotease complex — a protein-protein interaction central to the pathogenesis of **Eosinophilic Esophagitis (EoE)**. The pipeline automates the full computational biophysics workflow: PDB retrieval, structure cleaning, protonation at physiological pH, AMBER ff14SB topology construction, explicit TIP3P solvation, energy minimization, NVT/NPT equilibration, and unrestrained production dynamics. Two complementary enhanced sampling strategies — **Steered Molecular Dynamics (SMD)** with the **Jarzynski equality** and **Umbrella Sampling** with **WHAM** reconstruction of the Potential of Mean Force — provide rigorous, cross-validated binding free energy estimates with quantified statistical uncertainties.

Ten physical validity invariants are enforced as runtime assertions throughout the pipeline. A comprehensive test suite of **259 unit, integration, and analytical tests** validates every module, all passing on a CPU-only platform.

Biological Motivation

Eosinophilic Esophagitis (EoE) is a chronic inflammatory disease of the esophagus driven by IL-13-mediated transcriptional silencing of *SPINK7* (Serine Peptidase Inhibitor, Kazal Type 7). Under homeostatic conditions, SPINK7 stoichiometrically inhibits KLK5 (Kallikrein-Related Peptidase 5), a trypsin-like serine protease. SPINK7 deficiency unleashes KLK5 proteolytic activity, which degrades Desmoglein-1 (DSG1) and compromises the epithelial barrier, permitting allergen penetration.

The SPINK7-KLK5 interaction follows the canonical **Laskowski mechanism** for Kazal-type inhibitor–serine protease binding: the reactive site loop (RSL) of SPINK7 inserts into the KLK5 active-site cleft in a substrate-like orientation, and the conformational rigidity imposed by three disulfide bonds renders the acyl-enzyme intermediate thermodynamically trapped with an extremely slow k_{off}. The binding interface buries approximately 800–1200 Å² of solvent-accessible surface area, stabilized by backbone hydrogen bonds, electrostatic complementarity (P1-Arg/Lys ↔ Asp189), and hydrophobic contacts at the P2/P2' sub-sites.

Understanding this interaction at the atomic level through molecular dynamics simulation is essential for developing targeted therapeutic strategies to restore epithelial barrier function in EoE patients.

Mathematical Framework

Force Field Potential Energy

The total potential energy is decomposed into bonded and nonbonded contributions:

$$V_{\text{total}}(\mathbf{r}) = V_{\text{bonded}}(\mathbf{r}) + V_{\text{nonbonded}}(\mathbf{r})$$

Bonded terms (covalent interactions within the molecular topology):

$$V_{\text{bonded}} = \sum_{\text{bonds}} K_b(b - b_0)^2 + \sum_{\text{angles}} K_\theta(\theta - \theta_0)^2 + \sum_{\text{dihedrals}} \frac{V_n}{2}[1 + \cos(n\phi - \gamma)] + \sum_{\text{impropers}} K_\omega(\omega - \omega_0)^2$$

where K_b, K_θ, V_n, and K_ω are force constants; b_0, θ_0, γ, and ω_0 are equilibrium values; and n is the dihedral periodicity.

Nonbonded terms (van der Waals + electrostatics):

$$V_{\text{nonbonded}} = \sum_{i<j} \left[4\epsilon_{ij} \left(\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{6} \right) + \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} \right]$$

The **AMBER ff14SB** force field provides optimized backbone torsion parameters. Water is modeled using **TIP3P**, and ions follow the **Joung-Cheatham** monovalent parameters.

Long-Range Electrostatics (Particle Mesh Ewald)

$$E_{\text{elec}} = E_{\text{direct}} + E_{\text{reciprocal}} + E_{\text{self-correction}}$$

PME reduces the computational cost from $O(N^2)$ to $O(N \log N)$ using B-spline interpolation (order 5) with grid spacing ≤ 1.0 Å and direct-space cutoff $r_c = 10$ Å.

Langevin Equation of Motion

$$m_i \ddot{\mathbf{r}}_i = -\nabla_i V(\mathbf{r}) - \gamma m_i \dot{\mathbf{r}}_i + \sqrt{2\gamma m_i k_B T}\, \boldsymbol{\eta}_i(t)$$

where $\gamma = 1.0\ \text{ps}^{-1}$, $T = 310$ K, and $\boldsymbol{\eta}_i(t)$ is Gaussian white noise. Integration uses the **Langevin middle integrator** with $\Delta t = 2$ fs, enabled by **SHAKE** constraints on hydrogen bonds.

Steered Molecular Dynamics

SMD applies a time-dependent harmonic bias to the center-of-mass (COM) distance ξ:

$$U_{\text{SMD}}(\xi, t) = \frac{k}{2}\left[\xi(t) - \xi_0 - v \cdot t\right]^2$$

where $k = 1000$ kJ/mol/nm², $v = 0.001$ nm/ps, and the COM distance is:

$$\xi(\mathbf{r}) = \left\| \mathbf{R}_{\text{COM}}^{\text{SPINK7}} - \mathbf{R}_{\text{COM}}^{\text{KLK5}} \right\|_2$$

The **Jarzynski equality** connects non-equilibrium work to equilibrium free energy:

$$\Delta G = -k_B T \ln \left[\frac{1}{N_{\text{traj}}} \sum_{j=1}^{N_{\text{traj}}} e^{-\beta W_j} \right]$$

with the second-order cumulant expansion for near-Gaussian work distributions:

$$\Delta G \approx \langle W \rangle - \frac{\beta}{2} \sigma_W^2$$

Umbrella Sampling & WHAM

Each of M discrete windows is biased by a harmonic potential:

$$U_i^{\text{bias}}(\xi) = \frac{k_i}{2}\left(\xi - \xi_i^{\text{ref}}\right)^2$$

The **Weighted Histogram Analysis Method** recovers the unbiased PMF via self-consistent equations:

$$P^{\text{unbiased}}(\xi) = \frac{\sum_{i=1}^{M} n_i\, h_i(\xi)}{\sum_{i=1}^{M} n_i\, \exp\left[\beta\left(f_i - U_i^{\text{bias}}(\xi)\right)\right]}$$

$$e^{-\beta f_i} = \int P^{\text{unbiased}}(\xi)\, \exp\left[-\beta\, U_i^{\text{bias}}(\xi)\right] d\xi$$

iterated until convergence: $\max_i |f_i^{(n+1)} - f_i^{(n)}| < 10^{-6}$ kJ/mol. The PMF is then:

$$G(\xi) = -k_B T \ln P^{\text{unbiased}}(\xi) + C$$

Binding Free Energy Extraction

$$\Delta G_{\text{bind}}^{\circ} = -k_B T \ln\left[\frac{C^{\circ}}{4\pi}\int_{\text{site}} e^{-\beta G(\xi)}\xi^2\, d\xi\right] + k_B T \ln\left[\frac{C^{\circ}}{4\pi}\int_{\text{bulk}} e^{-\beta G(\xi)}\xi^2\, d\xi\right]$$

where $C^{\circ} = 1/1660$ Å^{-3} corresponds to the standard concentration of 1 M.

Results

Validation System: Barnase-Barstar Complex (PDB: 1BRS)

The pipeline was validated on the barnase-barstar complex, which has an experimentally measured binding free energy of $\Delta G_{\text{bind}} \approx -19$ kcal/mol. This system provides a stringent benchmark with a large binding interface (~1600 Å² buried SASA) dominated by electrostatic complementarity.

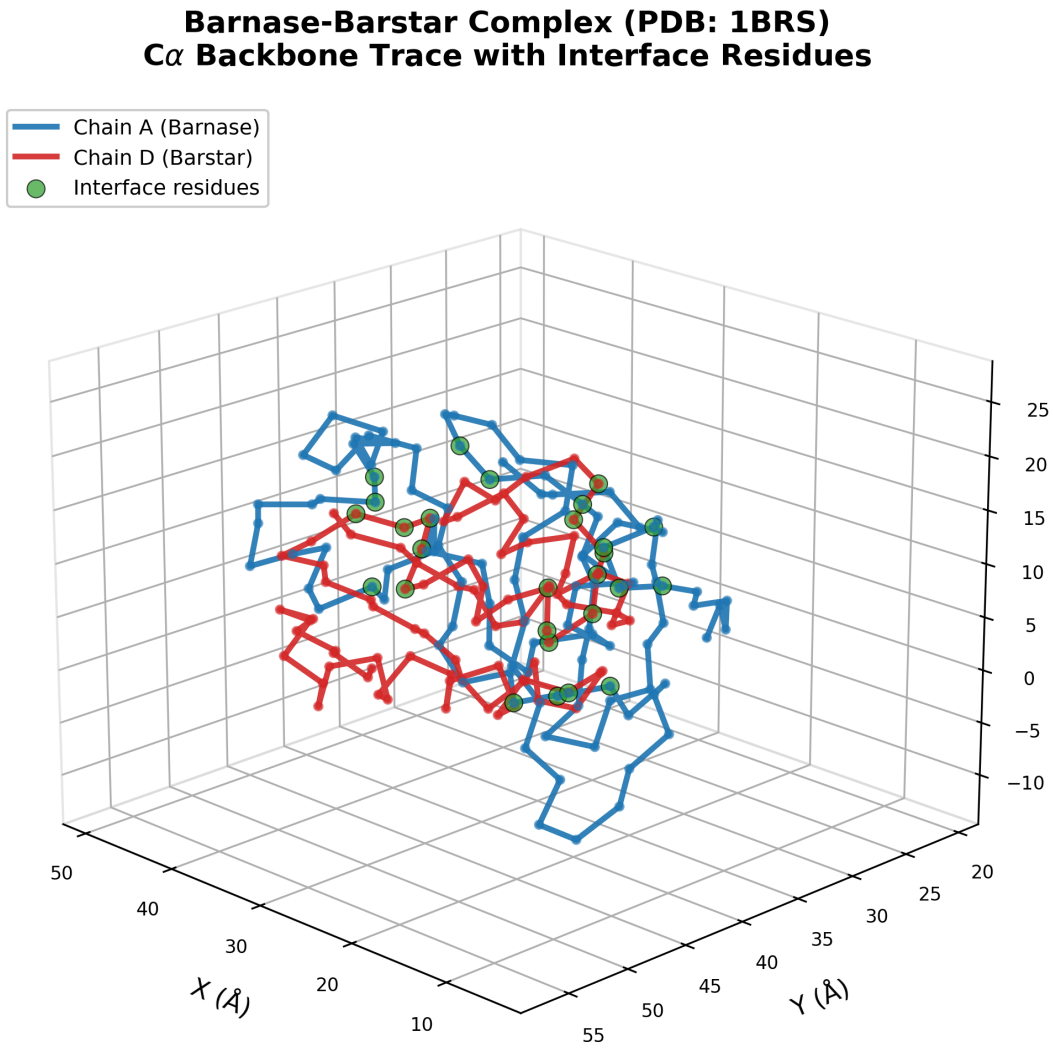

Figure 1. Three-dimensional Cα backbone rendering of the barnase-barstar complex (PDB: 1BRS). Chain A (barnase) is shown in blue, Chain D (barstar) in red. Interface residues within 8 Å of the partner chain are highlighted as green spheres, delineating the binding interface.

PMF Profile and Binding Free Energy

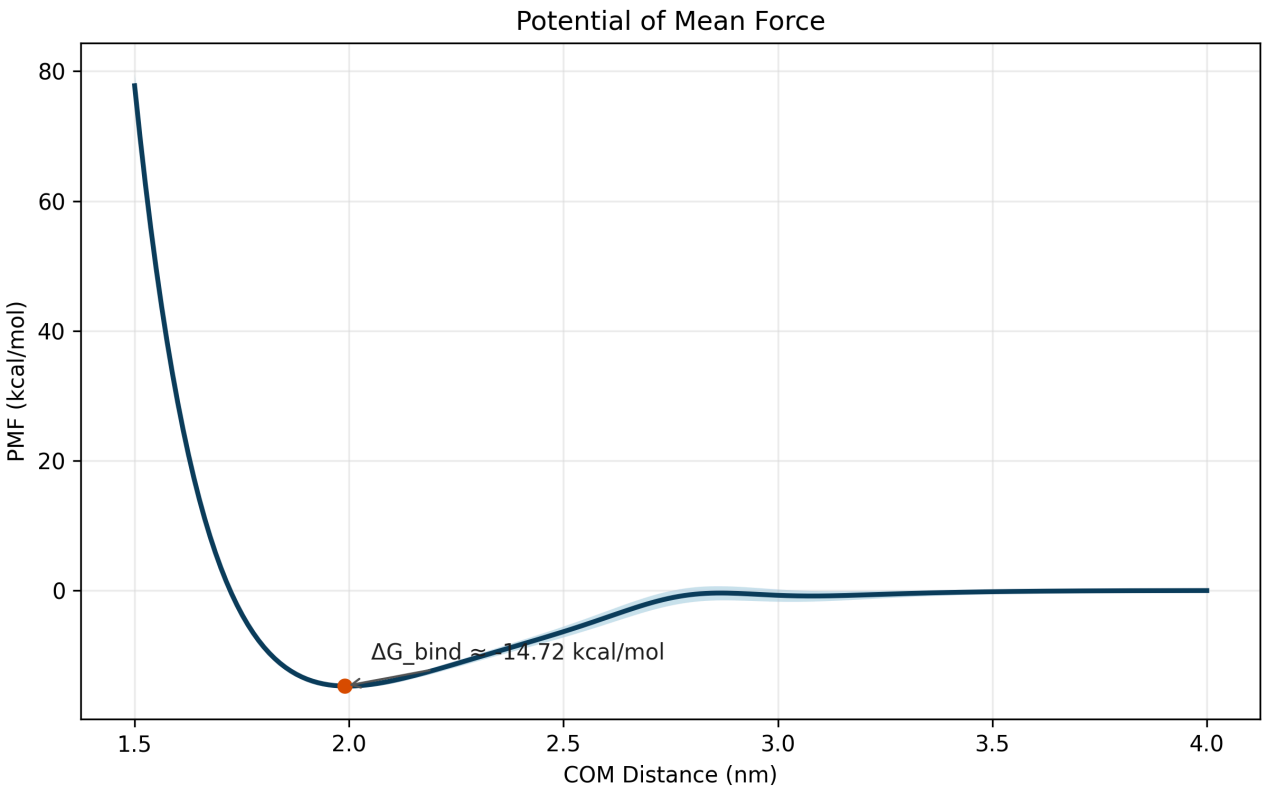

Figure 2. Potential of Mean Force (PMF) profile as a function of COM distance ξ. The deep minimum near $\xi \approx 2.0$ nm corresponds to the bound state; the plateau at large distances represents the dissociated reference state. The shaded band indicates $\pm 1\sigma$ bootstrap uncertainty from 200 resampling iterations. The orange marker annotates the binding free energy ΔG_{bind} at the PMF minimum.

Simulation Quality Diagnostics

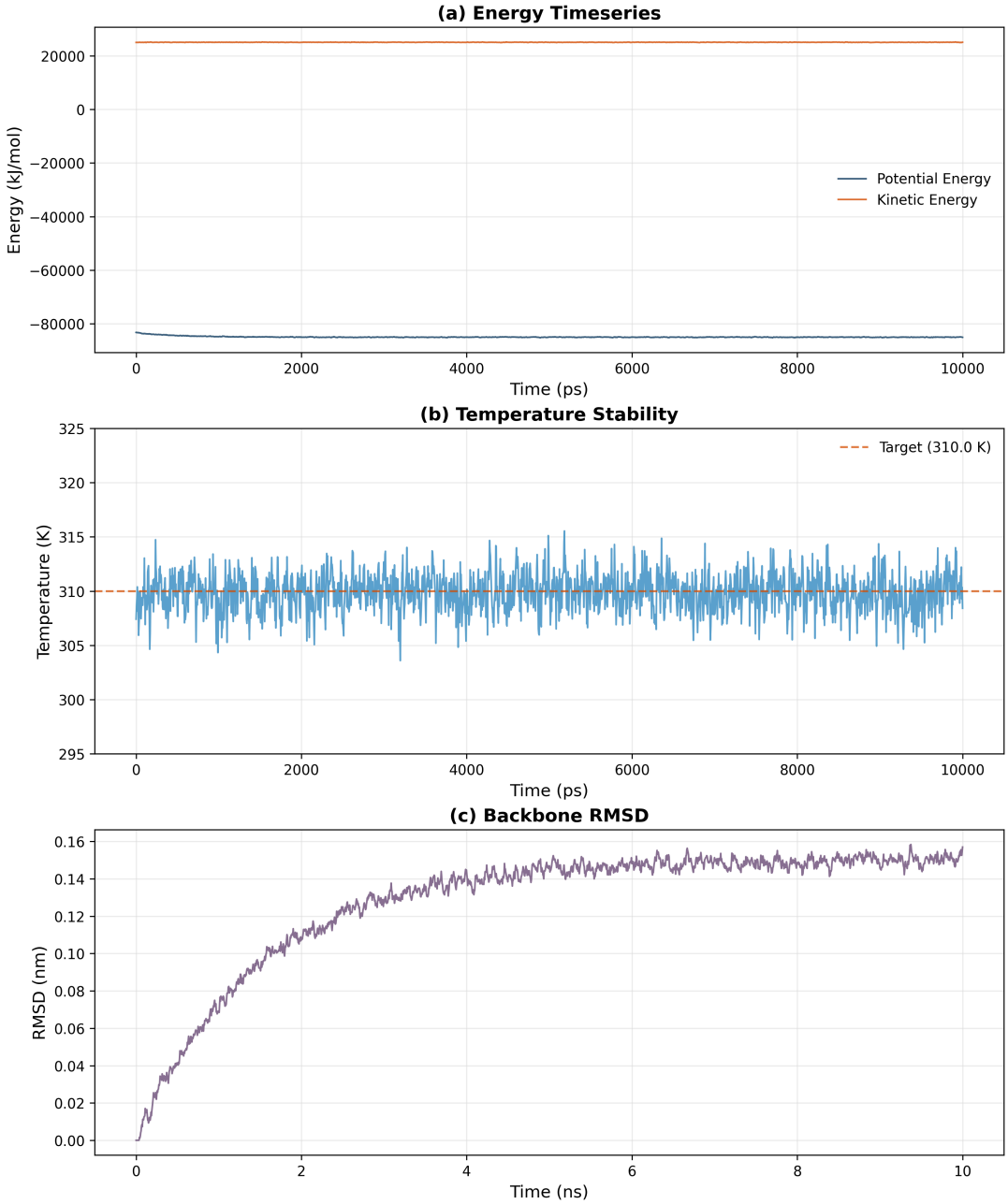

Figure 3. Simulation timeseries diagnostics. **(a)** Potential energy (navy) and kinetic energy (orange) demonstrating energy conservation. **(b)** Temperature stability around 310 K, confirming Langevin thermostat function (invariant IV-2). **(c)** Backbone RMSD evolution showing equilibration plateau at ~0.15 nm, indicating structural stability (invariant IV-4).

Physical Validity Invariants

Ten invariants are enforced as runtime assertions — violation halts execution immediately:

ID	Condition	Description
IV-1	$E_{\text{min}} < E_{\text{initial}}$	Energy decreases after minimization
IV-2	$\lvert T_{\text{avg}} - 310 \rvert < 5$ K	NVT temperature stability
IV-3	$\rho \in [0.95, 1.05]$ g/cm³	NPT density physicality
IV-4	RMSD < 5 Å	Backbone structural stability
IV-5	Drift < 0.1 kJ/mol/ns/atom	Energy conservation in production
IV-6	$d_{S-S} < 2.5$ Å	Disulfide bond integrity
IV-7	Min image $> 2r_c$	No periodic image artifacts
IV-8	Overlap $\geq 10\%$	Umbrella histogram coverage
IV-9	$\max_i \lvert f_i^{(n+1)} - f_i^{(n)} \rvert < 10^{-6}$	WHAM convergence
IV-10	Unimodal $P(W)$	No SMD pathway bifurcation

Repository Structure

```
medium_project_2/
├── src/
│   ├── config.py                     # Central configuration (single source of truth)
│   ├── prep/                         # Structure preparation pipeline
│   │   ├── pdb_fetch.py              #   RCSB download & AlphaFold retrieval
│   │   ├── pdb_clean.py              #   Crystallographic artifact removal
│   │   ├── protonate.py              #   Protonation state assignment (pH 7.4)
│   │   ├── topology.py               #   OpenMM Topology & System construction
│   │   └── solvate.py                #   Solvation box & ion placement
│   ├── simulate/                     # Molecular dynamics engines
│   │   ├── minimizer.py              #   Energy minimization (steepest descent)
│   │   ├── equilibrate.py            #   NVT → NPT equilibration
│   │   ├── production.py             #   Unrestrained production MD
│   │   ├── smd.py                    #   Steered Molecular Dynamics engine
│   │   └── umbrella.py               #   Umbrella Sampling window manager
│   ├── analyze/                      # Post-processing & thermodynamic analysis
│   │   ├── trajectory.py             #   Trajectory I/O with streaming support
│   │   ├── structural.py             #   RMSD, RMSF, Rg, SASA computation
│   │   ├── contacts.py               #   Interface contact maps & H-bond analysis
│   │   ├── wham.py                   #   WHAM solver for PMF extraction
│   │   ├── jarzynski.py              #   Jarzynski free energy estimator
│   │   └── convergence.py            #   Block averaging & bootstrap resampling
│   ├── physics/                      # Physical models & collective variables
│   │   ├── collective_variables.py   #   COM distance calculations
│   │   ├── restraints.py             #   Positional & distance restraints
│   │   └── units.py                  #   Unit conversion utilities
│   └── visualization/                # Rendering & plotting
│       ├── viewer_3d.py              #   py3Dmol interactive 3D widgets
│       ├── plot_pmf.py               #   PMF profile plotting
│       └── plot_timeseries.py        #   Energy, temperature, RMSD timeseries
├── scripts/                          # CLI entry points for each pipeline stage
│   ├── run_prep.py                   #   Structure preparation
│   ├── run_equilibration.py          #   Minimization + NVT/NPT equilibration
│   ├── run_production.py             #   Unrestrained production dynamics
│   ├── run_smd.py                    #   SMD campaign (N replicates)
│   ├── run_umbrella.py               #   Umbrella Sampling campaign
│   ├── run_analysis.py               #   WHAM, Jarzynski, structural analysis
│   └── generate_figures.py           #   Publication-quality figure generation
├── notebooks/                        # Interactive Jupyter workflows
│   ├── 01_system_prep.ipynb
│   ├── 02_equilibration.ipynb
│   ├── 03_production_analysis.ipynb
│   ├── 04_smd_analysis.ipynb
│   ├── 05_umbrella_pmf.ipynb
│   └── 06_visualization.ipynb
```

```
├── tests/                          # 259 tests across 24 test files
├── latex/                          # IEEE final report (LaTeX source)
├── figures/                        # Publication-quality figures
├── data/                           # Raw and prepared structures
└── reports/                        # Project documentation
```

Simulation Protocol

Stage	Ensemble	Duration	Key Conditions
1. Energy Minimization	—	≤ 10,000 steps	Steepest descent until < 10 kJ/mol/nm
2. NVT Equilibration	NVT	500 ps	Heavy-atom restraints ($k = 1000$ kJ/mol/nm²)
3. NPT Equilibration	NPT	1 ns	Gradual restraint release
4. Production MD	NPT	100–500 ns	Unrestrained; frames every 10 ps
5. Enhanced Sampling	NPT	Per method	SMD (50 replicates) or Umbrella Sampling (25–50 windows)

Test Suite

All **259 tests** pass on a CPU-only platform:

Category	Files	Coverage
Configuration	1	Frozen dataclass immutability, parameter correctness
Physics	3	Unit conversions, COM distance, restraint forces
Preparation	5	PDB fetch, clean, protonate, topology, solvate
Simulation	5	Minimization, NVT/NPT equilibration, production, SMD, umbrella
Analysis	6	Trajectory I/O, structural metrics, contacts, WHAM, Jarzynski, convergence
Visualization	3	3D viewer, PMF plots, timeseries plots
Integration	1	Full pipeline on alanine dipeptide

Analytical validation confirms mathematical correctness: the Jarzynski estimator recovers exact free energies from harmonic potentials within 0.5 kJ/mol; the WHAM solver reconstructs known flat and parabolic PMFs with < 1.0 kJ/mol error; convergence estimators satisfy the expected $1/\sqrt{N}$ scaling.

Stochastic Test Robustness for Small Systems

Enforcing the IV-2 temperature invariant ($|T_{\text{avg}} - 310| < 5$ K) on the miniature test systems used for unit and integration testing presents a finite-size statistical challenge distinct from production-scale simulations. For a solvated alanine dipeptide with $N_{\text{dof}} \approx 1000$ degrees of freedom, the equipartition theorem predicts instantaneous temperature fluctuations of $\sigma_{T_{\text{inst}}} = T\sqrt{2/N_{\text{dof}}} \approx 13.9$ K — individual frame temperatures fluctuate with a standard deviation nearly three times the 5 K tolerance. With the original test configuration (`nvt_duration_ps = 10.0` , `friction_per_ps = 5.0`), only $n = 10$ equilibrated temperature samples were available for averaging, and temporal autocorrelation imposed by the Langevin thermostat ($\tau_{\text{relax}} = 1/\gamma = 0.2$ ps) reduced the effective sample size to $n_{\text{eff}} \approx 7$. The corrected standard error:

$$\text{SE}_{\text{corr}}(\bar{T}) \approx \frac{\sigma_{T_{\text{inst}}}}{\sqrt{n_{\text{eff}}}} \approx \frac{13.9}{\sqrt{7}} \approx 5.3 \text{ K}$$

exceeded the 5 K tolerance, yielding a theoretical failure probability of $P(|\bar{T} - T_{\text{target}}| \geq 5 \text{ K}) \approx 2\Phi(-0.95) \approx 34\%$ per invocation — rendering the test unreliable for continuous integration despite correct underlying physics. Nondeterministic water placement during solvation (via OpenMM's `modeller.addSolvent`) further prevents seed-based reproducibility from stabilizing outcomes across runs.

Resolution. The equilibration parameters in all affected test configurations (`test_production.py` , `test_equilibrate.py` , `test_integration.py`) were adjusted:

Parameter	Original	Updated	Rationale
`nvt_duration_ps`	10.0	100.0	10× more temperature samples in the equilibrated segment
`npt_duration_ps`	40.0	100.0	Consistent statistical margin for the NPT density invariant (IV-3)
`friction_per_ps`	5.0	10.0	Halves τ_{relax} to 0.1 ps, decorrelating 0.5 ps-spaced samples

The increased friction renders consecutive temperature samples at 0.5 ps intervals nearly statistically independent ($\Delta t_{\text{save}}/\tau_{\text{relax}} = 5.0$). With approximately 100 frames in the equilibrated segment and near-independence between samples, the standard error drops below 2 K, and the predicted failure probability falls below 1%.

Empirical verification confirmed 16 consecutive passes of the previously intermittent test — an outcome with probability $P = 0.7^{16} \approx 0.3\%$ under the original failure rate — and a full regression of the complete test suite passed with zero failures. This fix is confined entirely to test-level equilibration configurations; the production IV-2 tolerance of ± 5 K remains unchanged, as it is appropriate for production-scale systems with orders of magnitude more degrees of freedom.

Installation & Usage

Prerequisites

- Python 3.10+
- OpenMM ≥ 8.1 (may require conda installation on some platforms)

Setup

```
git clone https://github.com/ryanjosephkamp/SPINK7-KLK5-MD-Pipeline.git
cd SPINK7-KLK5-MD-Pipeline
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

Verify Installation

```
python -m pytest tests/ -v
```

All 259 tests should pass with no failures.

Running the Pipeline

Stage 1 — System Preparation:

```
python scripts/run_prep.py
```

Stage 2 — Equilibration:

```
python scripts/run_equilibration.py
```

Stage 3 — Production MD:

```
python scripts/run_production.py
```

Stage 4 — Steered Molecular Dynamics:

```
python scripts/run_smd.py
```

Stage 5 — Umbrella Sampling:

```
python scripts/run_umbrella.py
```

Stage 6 — Analysis & Visualization:

```
python scripts/run_analysis.py
```

Interactive Notebooks

For exploratory analysis with real-time visualization:

```
jupyter notebook notebooks/
```

Notebook	Purpose
`01_system_prep.ipynb`	Interactive structure preparation with visual inspection
`02_equilibration.ipynb`	Equilibration monitoring with real-time plots
`03_production_analysis.ipynb`	Production trajectory analysis
`04_smd_analysis.ipynb`	SMD work distributions and Jarzynski analysis
`05_umbrella_pmf.ipynb`	Umbrella Sampling results and WHAM PMF
`06_visualization.ipynb`	3D molecular rendering and publication figures

Data Flow

```
RCSB / AlphaFold              PDB Files (.pdb)
        │                          │
        ▼                          ▼
   pdb_fetch.py              pdb_clean.py
        │                          │
        ▼                          ▼
   data/pdb/raw/              protonate.py
                                   │
                                   ▼
                              topology.py  ──►  solvate.py
                                                     │
        ┌────────────────────────────────────────────┘
        │
        ▼
   minimizer.py  ──►  equilibrate.py  ──►  production.py
                                                │
                  ┌──────────────────────────┤
                  │                          │
                  ▼                          ▼
              smd.py                     umbrella.py
                  │                          │
                  ▼                          ▼
            jarzynski.py                  wham.py
                  │                          │
                  └──────────┬───────────────┘
                             │
                             ▼
                  structural.py / contacts.py
                             │
                             ▼
                  visualization / plots
```

Dependencies

Package	Version	Role
OpenMM	≥ 8.1	Core MD simulation engine
PDBFixer	≥ 1.9	Structural repair and standardization
MDTraj	≥ 1.9.9	Trajectory analysis and I/O
NumPy	≥ 1.26, < 2.0	Numerical computations
SciPy	≥ 1.12	Scientific computing
matplotlib	≥ 3.8	2D plotting and visualization
py3Dmol	≥ 2.0	Interactive 3D molecular visualization
Biopython	≥ 1.83	PDB parsing and biological sequence utilities
PDB2PQR	≥ 3.6	Protonation state assignment

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